

# Mulheres at IMPA: fluid dynamics session

<sup>1</sup> IMPA, Instituto de Matematica  
Pura e Aplicada, Rio de Janeiro, Brazil

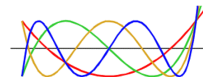


Yulia Petrova<sup>1,2</sup>

<https://yulia-petrova.github.io/>

12 May 2022

<sup>2</sup> St Petersburg State University,  
Chebyshev Lab, Russia



## Two words about me...

- I grew up in a small city Ukhta in the north of Russia,  $\approx 100$  thousand citizens
- At school participated and sometimes won various town and republic Olympiads: maths, physics, chemistry, informatics
- At 17 years moved to St Petersburg, Russia, to study at University



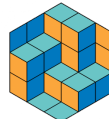
## Professional experience

2008–2013 | MSc in Mathematics, SPbSU, Russia.  
**Department of PDEs**

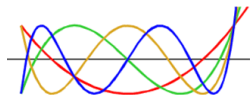
2013–2018 | PhD, SPbSU, Russia. “Exact  $L_2$ -small ball asymptotics for finite-dimensional perturbations of Gaussian processes”.  
**Probability theory & spectral theory**

2017–2021 | Researcher at Chebyshev Laboratory, SPbSU, Russia. Participant of industrial projects with petroleum company “Gazprom-Neft”. **Fluid dynamics**

2021–2023 | Postdoc at IMPA. **Fluid dynamics.**  
Working with Prof. Dan Marchesin.



St Petersburg  
University



# Great women in maths from Russia



Sofia Kovalevskaya



Olga Ladyzhenskaya



Nina Uraltseva



Olga Oleinik

NB: biographies of women mathematicians:

<https://mathwomen.agnesscott.org/women/women.htm>

# Motion of fluids inspires..!

- Euler equation (1755):  
ideal fluid

$$\begin{aligned}\partial_t u + (u \cdot \nabla)u &= \nabla p, \\ \nabla \cdot u &= 0.\end{aligned}$$

- Navier-Stokes equation (1845):  
adds viscosity

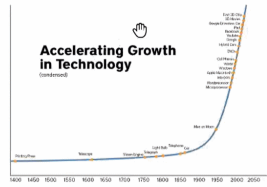
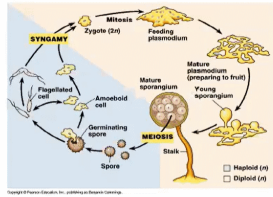
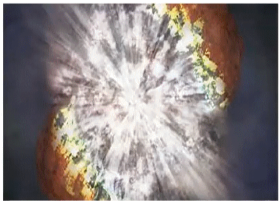
$$\partial_t u + (u \cdot \nabla)u - \nu \Delta u = \nabla p$$



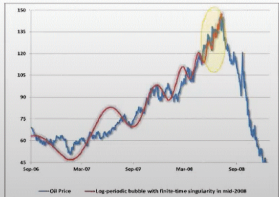
See more photos: Van Dyke “Album of fluid motion”

# Fluid dynamics (PDEs) often deal with “singularities”

Dramatic phenomena. Explosion, catastrophe, abrupt change.

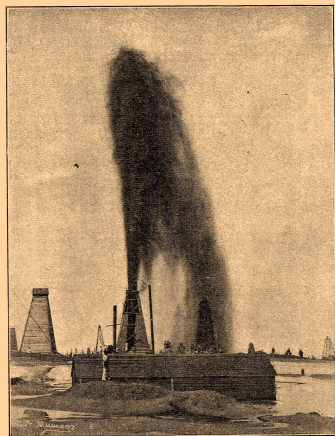


A shock wave, a transition from laminar flow to turbulence, or stock market crashes can also be linked to singularities in modelling equations.



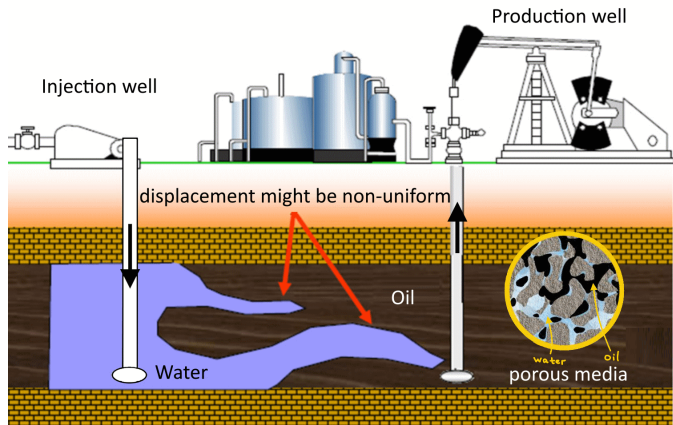
# I am interested in mathematics of oil recovery

How oil was recovered in the beginning? (Baku, 1857)

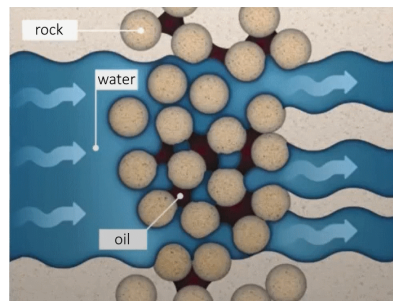
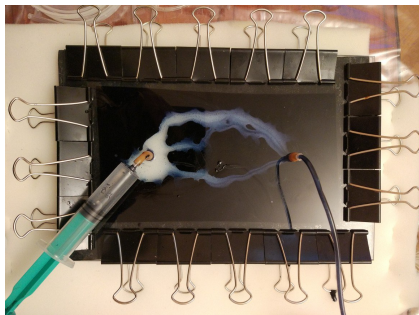


Нефтяной фонтан Торгового товарищества, бывший в центре в 1857 г. (Благодаря у Баксу).

How oil is recovered now?



# Problems: macroscopic and microscopic sweep efficiency



- happens due to very viscous oil or inhomogeneous media
- local entrapment of oil in pores due to high capillary pressure

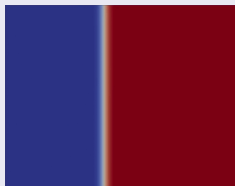
## Possible solution

- Inject gas ( $\text{CO}_2$ , natural) to decrease the oil viscosity
- Add chemicals (polymer) to increase the water viscosity
- Add chemicals (surfactant) that reduce the surface tension etc

# Fundamental research: two main directions

## 1-dim in spatial variable

- Stable displacement



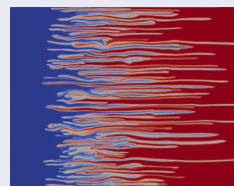
- main question: find an exact solution to a Riemann problem
- hyperbolic conservation laws

$$\begin{aligned} s_t + f(s, c)_x &= 0, \\ (cs + a(c))_t + (cf(s, c))_x &= 0. \end{aligned}$$

Example: chemical flooding model

## 2-dim (or 3-dim) in spatial variable

- Unstable displacement



- source of instability: water and oil/polymer have different viscosities
- viscous fingering phenomenon

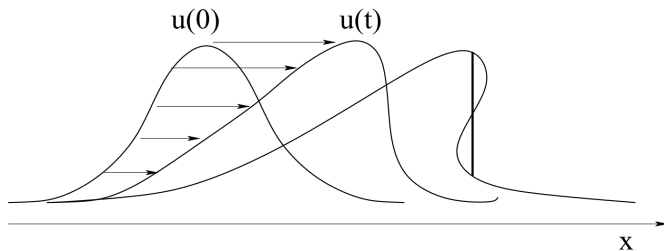
$$\begin{aligned} c_t + u \cdot \nabla c &= \varepsilon \Delta c, \\ \operatorname{div}(u) &= 0, \\ u &= -\nabla p / \mu(c). \end{aligned}$$

Example: Peaceman model

# Hyperbolic systems of conservation laws

$$u_t + \left( \frac{u^2}{2} \right)_x = 0 \quad (\text{Burger's equation, 1948})$$

- non-linearity implies wave speed  $\lambda(u) = u$  depends on state  $u$
- So the wave can spread (rarefaction wave) or concentrate (shock wave)



$$u_t + (f(u))_x = 0 \quad (\text{Buckley-Leverett equation, water flooding})$$

- existence, uniqueness was established by Olga Oleinik (1957)

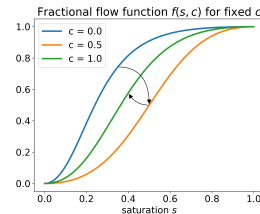
# Problem statement

Chemical flooding can be described as the system of conservation laws ( $x \in \mathbb{R}, t > 0$ ):

$$s_t + f(s, c)_x = 0, \quad (\text{conservation of water}) \quad (1)$$

$$(cs + a(c))_t + (cf(s, c))_x = 0. \quad (\text{conservation of chemical})$$

- $s = s(x, t)$  — water phase saturation;
- $f(s, c)$  — fractional flow function (usually  $S$ -shaped);
- $c = c(x, t)$  — concentration of a chemical agent in water;
- $a(c)$  — adsorption of a chemical agent on a rock (usually increasing, concave).



Initial data:

$$(s, c)|_{t=0} = \begin{cases} (1, 1), & \text{if } x \leq 0, \\ (0, 0), & \text{if } x > 0, \end{cases} \quad (2)$$

Aim:

Find a solution to initial-value problem (1)–(2) when  $f$  depends non-monotonically on  $c$ .

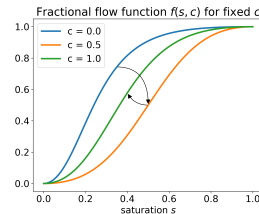
NB: Non-monotone dependence appears in surfactant flooding, low salinity water flooding etc

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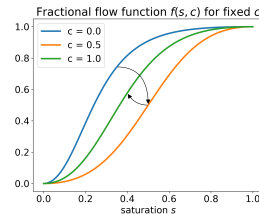
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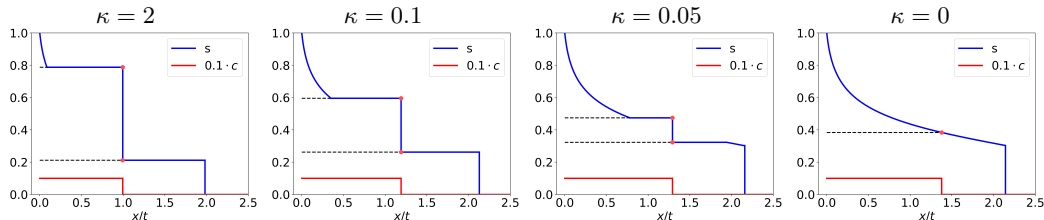
NB: Non-monotone dependence appears in surfactant flooding, low salinity water flooding etc

# Result

- weak solutions  $\Rightarrow$  non-uniqueness of solutions to a Riemann problem
- use vanishing viscosity criterion — add small diffusion/capillary terms

$$\begin{aligned} s_t + f(s, c)_x &= \varepsilon_c (A(s, c) s_x)_x, \\ (cs + a(c))_t + (cf(s, c))_x &= \varepsilon_c (cA(s, c) s_x)_x + \varepsilon_d c_{xx}. \end{aligned}$$

- $f(s, c)$  monotone in  $c \Rightarrow$  uniqueness of vanishing viscosity solution (1988)
- **Interesting idea:**  $f(s, c)$  non-monotone in  $c \Rightarrow$  **exist multiple** vanishing viscosity solutions, depending on ratio  $\kappa = \varepsilon_d / \varepsilon_c$



NB: the appeared shock is known as undercompressive (transitional).

# Undercompressive shocks

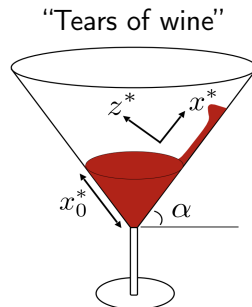
- Appear in various contexts:
  - three-phase flows
  - thin film theory
- depend heavily on the diffusion terms
- take solution in the form of a travelling wave

$$s = s(\xi) = s(x - vt) \quad c = c(\xi) = c(x - vt)$$

get the dynamical system on  $(s, c)$

$$\begin{aligned} s_\xi &= f(s, c) - v(s + d_1), \\ c_\xi &= v(d_1 c - d_2 - a(c))/\kappa. \end{aligned}$$

Undercompressive shocks correspond to saddle-to-saddle heteroclinic orbits



# Two-dimensional problems

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Show video

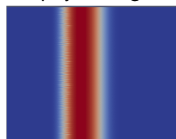
## 2d problems: motivation and main question

- water flooding;
- chemical flooding;
- cause problems for oil recovery.

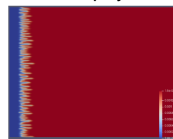
### Main questions of interest:

- 1 what are rigorous bounds on velocities of mixing zone propagation?
- 2 how to calculate the optimal size of the polymer slug?

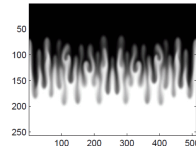
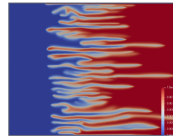
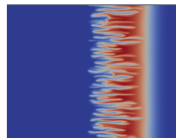
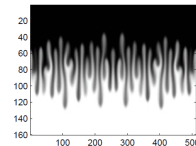
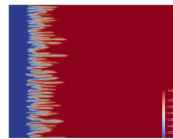
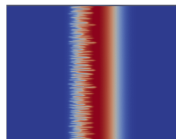
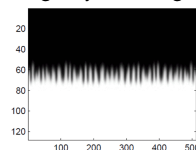
polymer slug



water into polymer



gravity-driven fingers



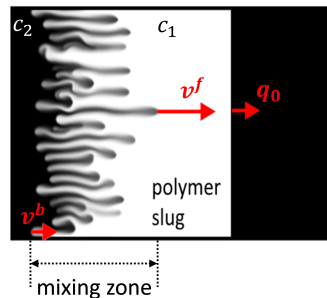
# Velocities of the mixing zone

One-phase miscible displacement  
(Peaceman model)

$$c_t + u \cdot \nabla c = \varepsilon \Delta c,$$

$$\operatorname{div}(u) = 0,$$

$$u = -\frac{\nabla p}{\mu(c)}.$$



Numerically and experimentally we observe:

- $q_0$  — velocity of the stable front
- $v_f$  — velocity of the front end of the mixing zone **is constant**
- $v_b$  — velocity of the back end of the mixing zone **is constant**

Open problem: rigorously prove the linear growth of the mixing zone

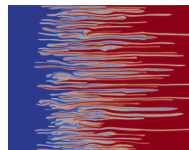
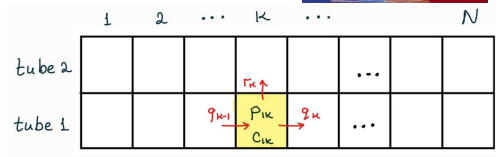
# Toy model of viscous fingering (work in progress)

## Discrete case

- system of  $2N$  ODEs and  $N$  algebraic equations

## Unknowns:

- $c_{1k}(t), c_{2k}(t)$  — concentrations
- $p_{1k}, p_{2k}$  — pressures
- $q_k(t), r_k(t)$  — velocities

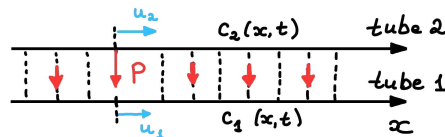


## Continuous case

- two coupled advection-diffusion eqs

$$\partial_t c_1 = -\partial_x(u_1 c_1) + \partial_x u_{1,2} \cdot c_{1,2} + \varepsilon \partial_{xx} c_1,$$

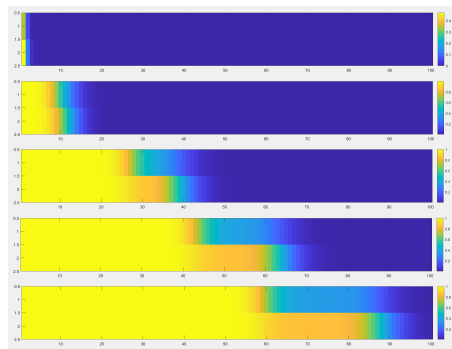
$$\partial_t c_2 = -\partial_x(u_2 c_2) - \partial_x u_{1,2} \cdot c_{1,2} + \varepsilon \partial_{xx} c_2.$$



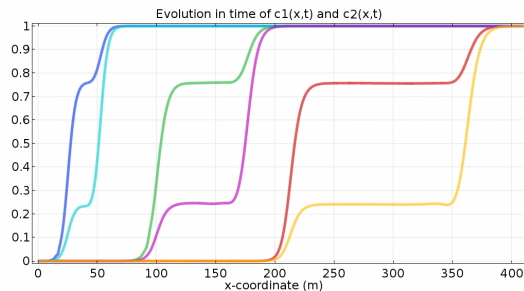
Work in progress with S. Tikhomirov, Y. Efendiev.

# Toy model of viscous fingering: numerical experiments

## Discrete setting



## Continuous setting



Result of experiments: cascade of two travelling waves (TW)

$$(0, 0) \xrightarrow{\text{TW}_1} (c_1^*, c_2^*) \xrightarrow{\text{TW}_2} (1, 1)$$

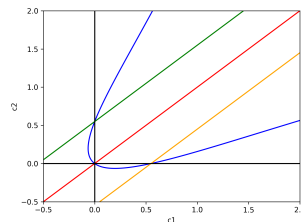
# Toy model of viscous fingering: theoretical approach

For a travelling wave  $c_1 = c_1(x - vt)$  and  $c_2 = c_2(x - vt)$  we have a dynamical system.

## Travelling wave dynamical system

$$\begin{aligned}c_1' &= g, \\g' &= (u - v)g, \\c_2' &= (u - v)c_1 + (2 - u - v)c_2 - g.\end{aligned}$$

Here  $u = u(c_1, c_2)$ , depends on  $\mu(c_1)$  and  $\mu(c_2)$ .



## Want to prove

For all  $v \in \mathbb{R}_+$  there exists a unique point  $(c_1^*, c_2^*)$ : there exists a trajectory  $U = U(\xi)$ :

$$U(-\infty) = (0, 0, 0) \quad U(+\infty) = (c_1^*, 0, c_2^*)$$

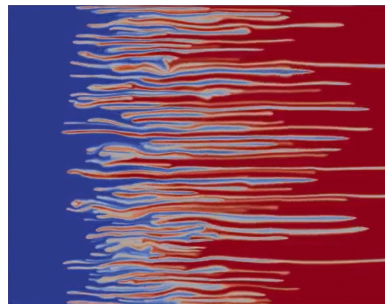
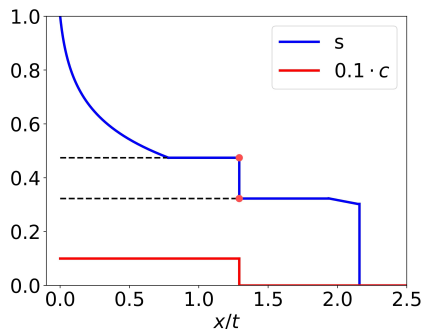
How to quantify the dependence  $c_1^* = c_1^*(v)$  and  $c_2^* = c_2^*(v)$ ?

Here  $U = (c_1, g, c_2)$ .

# Thank you for your attention!

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<https://yulia-petrova.github.io/>



# PEACE for Russia and Ukraine (and world)

These mathematicians will never prove a theorem because of the war...



Yuliia Zdanovska (Kiev)



Konstantin Olmezov (MIPT, Moscow)

Many Ukrainian mathematicians are under bomb attacks in Ukraine.  
Many Russian mathematicians are under political pressure in Russia.